

HOMEWORK #1 Maya Geva

1) Simplify as much as possible:

a) $0 + 0 + 1$

$= 1$ because both $0+1$ and $1+1 = 1$

b) $(y+x) \cdot 0$

$= 0$ because anything $\cdot 0$ is 0

c) $\underbrace{x+x}_{\text{becomes } x} + \underbrace{y+y}_{\text{becomes } y}$

$= x+y$

d) $1+1 = 1$

e) $A + \bar{A} = 1$ If $A=1 \rightarrow 1+0 = 1$
 $= 1$ If $A=0 \rightarrow 0+1 = 1$

2) a) "The train is not late and my watch is not fast"

b) $0 \leq x < 7$

is $p \oplus q \wedge r \equiv (p \wedge r) \oplus (q \wedge r)$?

3)

p	q	r	$p \oplus q \wedge r$	$(p \wedge r) \oplus (q \wedge r)$
T	T	T	F	F
T	T	F	F	F
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

the two statements are logically equivalent because they have the same truth table outcomes

$$(4) \sim(p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv \sim p$$

De Morgan's to distribute the $\sim$

$$\equiv (\sim p \wedge q) \vee (\sim p \wedge \sim q)$$

*distributive law $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$

$$\equiv \sim p \wedge (q \vee \sim q)$$

*negation law $p \vee \sim p \equiv t$

$$\equiv \sim p \wedge t$$

*identity law $p \wedge t \equiv p$

$$\equiv \sim p$$

$$\sim p \equiv \sim p \quad \checkmark$$

(5) "the converse & inverse of a statement are $\equiv$ "

original: $p \rightarrow q$, converse: $q \rightarrow p$, inverse: $\sim p \rightarrow \sim q$

		original converse inverse		
P	q	$p \rightarrow q$	$q \rightarrow p$	$\sim p \rightarrow \sim q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

the statement is

true b/c the converse

& inverse have the

same TT outcome

(6) "It is raining. It is not cold. It is cloudy."

p: It is raining

q: it is not cold

r: it is cloudy

"It is raining and it is cold, or it is not cloudy"

$$(p \wedge q) \vee \sim r$$

$$\equiv (\text{true} \wedge \text{false}) \vee \text{false}$$

$$\equiv \text{false} \wedge \text{false}$$

$$\equiv \text{false} \rightarrow \text{therefore the statement is false}$$

$$(7) (\neg A \wedge \neg B) \vee (\neg A \wedge B) \vee (A \wedge B) \rightarrow \neg A \vee B$$

distributive law to take out $\neg A$

$$= \neg A \wedge (\neg B \vee B) \vee (A \wedge B)$$

*negation law

$$= (\neg A \wedge 1) \vee (A \wedge B)$$

*identity law

$$= \neg A \vee (A \wedge B)$$

*distributive laws

$$= (\neg A \vee A) \wedge (\neg A \vee B)$$

*negation law

$$= 1 \wedge (\neg A \vee B)$$

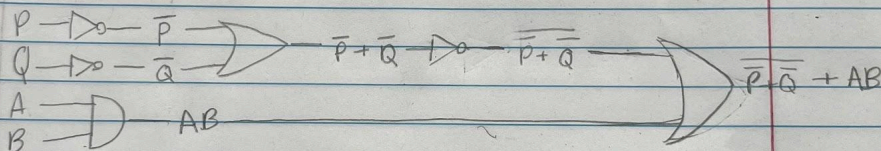
*identity law

$$= \neg A \vee B$$

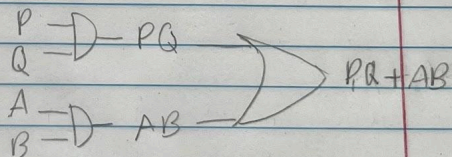
$$\neg A \vee B \equiv \neg A \wedge B \checkmark$$

$$(8) P + \bar{Q} + AB$$

a) draw circuit gates:



$$b) \bar{P} + \bar{Q} + AB = PQ + AB:$$



c) a) $A + B$: neither because it is sometimes T and sometimes F

b) $\neg B \wedge B \wedge \neg B$: according to the negation law, $\neg B \wedge B$ will always give false, and false $\wedge \neg B$ is also always false, so it's inconsistent

c) $\neg A \wedge (A \vee B) \wedge \neg B$:

All the truth table outcomes are false so it is inconsistent

A	B	$\neg A$	$\neg B$	$\neg A \wedge \neg B$	$A \vee B$	$\neg A \wedge (A \vee B) \wedge \neg B$
T	T	F	F	F	T	F
T	F	F	T	F	T	F
F	T	T	F	F	T	F
F	F	T	T	T	F	F

d) It wouldn't make sense to use a valid or inconsistent for computation because they only give one output (T or F). for computation as a circuit it seem like you want to use statements that will have a variety of outcomes based on the inputs